① A Sphenphered boy get bored tending the town's flock. To h[ave] some fun he cries out, "wolf!" even though no wolf in The Villages run to protect the flock, but then get reall[y] mad when they realize the boy was playing a Joke on the[m]

one night, the shepherd boys see a real wolf approaching the and call out "wolf!" The villages refuse to be fooled again and in their house. The hungry wolf turn the flock into ch[ops]. The town goes Hungry, panic ensure. identify (TP, F[P] ...

Sol

Actual \ predicted	Wolf = Yes	Wolf = NO
Wolf = Yes	waring = correct TP Wolf come = Yes	waring Not given FN wolf came
Wolf = NO	FP wolf come = NO	TN No warning No wolf,

② Assume there are 100 images, 30 of them depict a cat, the rest do no[t]. A occurence of machine learning model predict the occurence of cat 25 of 30 cat image. it also predict absence of a cat in 50 of 70 NO cat images identify TP, FP, FN, TN Calculate the Precision, Rec[all], F1, score

Sol

Cat = 30
Not = 70

Actual \ predicted	Cat = Yes	Cat = NO	
Cat = Yes	TP = 25	FN = 5	30
Cat = NO	FP 20	TN 50	70
	45	55	100

① precision = $\frac{Tp}{Tp+Fp}$ = $\frac{25}{25+20}$ = $\frac{25}{45}$ = 0.555 x100 = $\boxed{55.6}$

② Recall = $\frac{Tp}{Tp+FN}$ = $\frac{25}{25+5}$ = $\frac{25}{30}$ = 0.833 x 100 = $\boxed{83.3}$

③ F1 score = $\frac{2\times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}}$ = $\frac{2 \times 0.555 \times 0.833}{0.555+0.833}$

= $\frac{0.92463}{1.388}$ = 0.666 x100 = $\boxed{66.7}$

An email service provider want to classify email as spam or Not spam using features

- Contions "offers" (yes/NO)
- Contions "free" (yes/NO)
- Email Length (short/long)

Sample Dataset

offer	free	Length	class
yes	yes	short	spam
NO	NO	Long	NO spam
yes	yes	Long	spam
NO	NO	short	Not spam

① Calculate prior probabilties:

- P (spam), P (Not spam)

② Computer conditional probabilities for each features

③ using Naive Bayies classify

- (offer = yes, free = NO, length = short)

i) p(spam) = 3/6 = 1/2

p (Not spam) = 3/6 = 1/2

ii) Conditional probabilites

→ Spam

$\boxed{\text{Offer}}$

p (O = Y | spam) = $\frac{2}{3}$

p (O = No/spam = 1/3

$\boxed{\text{Free}}$

p (Free = y/spam) = 1

p (free = No/spam) = 0

Length

p(L = ~~yes~~ short /spam = 2/3

p(L = ~~No~~ long /spam = 1/3

Not spam

p = (O=y/Not) = 1/3

p = (O=N/Not spam) = 2/3 } Offer

p(Free = y/Not spam) = 0/3

p(free = N/Not spam) = 3/3 } Fr

p(Length = y/Not spam = 1/3

p (Length = N/ No Spam = 2/3

iii) Using Naive Bay's classify (Offer = yes , Free = No , Length = short)

p(O = Y , F=N, L=S)

Spam

$\frac{1}{2} \times 2/3 \times 0 \times \frac{2}{3} = 0$

= 1/2 × ... × ...

= p (spam) = 0

Not spam

$\frac{1}{2} \times 1/3 \times 1 \times \frac{1}{3}$

$= \frac{1}{18}$

p (Not) = 1/18

= 0.055

Prediction

P(spam) = 0

P(Not spam) = 0.056

Prediction will be No spam

ENTROPY

$$H(y) = -\sum_{i=1}^{m} p_i \log(p_i)$$

$$\text{Info}(D) = -\sum_{i=1}^{m} p_i \log_2(p_i)$$

$$\text{Info}_A(D) = \sum_{j=1}^{v} \frac{|D_j|}{|D|} \times \text{Info}(D_j)$$

$$\text{Gain}(A) = \text{Info}(D) - \text{Info}_A(D)$$

Assume,

class P: buy-computer = "yes"

class N: buy computer = "NO"

Table

Age	Class Yes P	Class No N	I(P,N)
<= 30	2	3	0.971
31-40	4	0	0
>40	3	2	0.971
	9	5	0.940

$I(2,3) = -\sum p_i \log(p_i)$

$= -\frac{2}{5}\log_2\left(\frac{2}{5}\right) - \frac{3}{5}\log_2\left(\frac{3}{5}\right)$

$= -\frac{2}{5}(-1.322) - \frac{3}{5}(-0.737)$

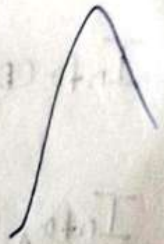
$= 0.5288 + 0.4422$

$= 0.971$

$I(4,0) = -\sum p_i \log(p_i)$

$= -\frac{4}{4}\log_2(1) - \frac{0}{4}\log_2\frac{(0)}{4}$

$= 0$



$I(3,2) = -\sum p_i \log(p_i)$

$= -\frac{3}{5}\log_2(3/5) - \frac{2}{5}\log_2(2/5)$

$= \frac{-3}{5}(-0.737) - \frac{2}{5}(-1.322)$

$= 0.4422 + 0.5286$

$= 0.971$

$I(9,5)$

$= -\sum p_i \log_2(p_i)$

$= \frac{-9}{14}\log_2\left(\frac{9}{14}\right) - \frac{5}{14}\log_2\left(\frac{5}{14}\right)$

$= \frac{-9}{14}(-0.6374) - \frac{5}{14}(-1.4854)$

$= 0.4097 + 0.5304$

$= 0.940$

Confusion matrix

Actual class / predicted class	Cancer = yes	Cancer = NO	Total
Cancer = yes	90	210	300 (P)
Cancer = NO	140	9560	9700 (N)
Total	230	9770	10000
	P'	N'	(All)

① Accuracy $= \frac{TP + TN}{All} = \frac{90 + 9560}{10000} = \frac{965\cancel{0}}{100\cancel{0}} = 0.965 \times 100$ $\boxed{96.5\%}$

② error rate $= 1 - \text{accuracy} = 1 - 96.5 = \boxed{3.5\%}$

③ Sensitivity $= \frac{\text{True positive}}{P} = \frac{90}{300} = 0.3 = \boxed{30\%}$

④ Specificity $= \frac{TN}{N} = \frac{9560}{9700} = 0.986 \times 100$

$= \boxed{98.6\%}$

⑤ precision $= \frac{TP}{TP + FP} = \frac{90}{90 + 140} = 0.391 \times 100 = \boxed{39.1\%}$

⑥ Recall $= \frac{TP}{TP + FN} = \frac{90}{90 + 210} = 0.3 \times 100 = \boxed{30\%}$

⑦ F1 score $= \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}}$

$\frac{2 \times 0.391 \times 0.3}{0.391 + 0.3} = \frac{0.2346}{0.691}$

$= 0.339 \times 100 = \boxed{33\%}$

$I(4,2) = -\frac{4}{6}\log_2\left(\frac{4}{6}\right) - \frac{2}{6}\log_2(2/6)$

$= -\frac{4}{6}(-0.585) - 2/6\,(-1.585)$

$= 0.389 + 0.528$

$= 0.917$

$I(3,1) = -\frac{3}{4}\log_2\left(\frac{3}{4}\right) - \frac{1}{4}\log_2\left(\frac{1}{4}\right)$

$= -\frac{3}{4}(-0.415) - \frac{1}{4}(-2)$

$= 0.311 + 0.5$

$= 0.811$

$Info_{income}(D) = \sum \frac{|D_j|}{|D|} \times info(D)$

$= \frac{4}{14} \times 0.811 + \frac{6}{14} \times 0.917 \times \frac{4}{14} \times 1$

$= \frac{0.231 + 0.285 + 0.392}{14}$

$= \frac{0.9081}{14}$

$Gain_{income} = 0.940 - 0.9081$

$\boxed{0.029}$

Gain (student) = 0.151

Gain (credit - rating) = 0.048

$$Info_{age}(D) = \sum \frac{|D_j|}{|D|} \times info(D_j)$$

$$= \frac{5}{14} \times 0.971 + \frac{4}{14} \times 0.0 + \frac{5}{14} \times 0.971 = 0.694$$

$$= 0.694$$

$$Gain(age) = 0.940 - 0.694$$

$$= 0.246$$

For the attribute income =

income	Yes P	No N	I(P,N)
high	2	2	1
medium	4	2	0.917
low	3	1	0.811
	9	5	0.940

$$I(2,2) = -\sum p_i \log(p_i)$$

$$= -\frac{2}{4}\log_2\left(\frac{2}{4}\right) - \frac{2}{4}\log_2\left(\frac{2}{4}\right)$$

$$= -\frac{2}{4}(-1) - \frac{2}{4}(-1)$$

$$= \frac{2}{4} + \frac{2}{4} = \frac{4}{4}$$

$$= 1$$

Gini index

$$\text{Gini}(D) = 1 - \sum_{j=1}^{n} p_j^2$$

the Gini index based on income

$$\text{Gini}(D) = 1 - \sum_{j=1}^{n} p_j^2$$

$$\text{Gini}(D) = 1 - \left[\left(\frac{9}{14}\right)^2 + \left(\frac{5}{14}\right)^2\right]$$

$$\left[\begin{array}{l} \left.\begin{array}{l} L - 4 \\ M - 6 \end{array}\right\} - 10 \\ H - 4 \rightarrow \frac{4}{14} \end{array}\right]$$

$$= 1 - [0.413 + 0.127]$$

$$= 1 - 0.54$$

$$= 0.46$$

partition of data set with respect to income

D_1 : (low, medium) -10

D_2 : (High) -4

$$\text{Gini}_{(income)}(D) = \frac{|D_1|}{|D|}\,\text{gini}(D_1) + \frac{|D_2|}{D}\,\text{gini}(D_2)$$

$$= \frac{10}{14}\left[1 - \left[\left(\frac{7}{10}\right)^2 + \left(\frac{3}{10}\right)^2\right]\right] + \frac{4}{14}\left[1 - \left[\left(\frac{2}{4}\right)^2 + \left(\frac{2}{4}\right)^2\right]\right]$$

$$= \frac{10}{14}\left[1 - [0.49] + [0.09]\right] + \frac{4}{14}\left[1 - [0.25 + 0.25]\right]$$

$$= \frac{10}{14}[1 - 0.58] + \frac{4}{14}[1 - 0.50]$$

$$= \frac{10}{14}[0.42] + \frac{4}{14}[0.50]$$

Gain ratio = gain (A) / Split info(A)

$$\text{Split info}_A(D) = -\sum_{j=1}^{v} \frac{|D_j|}{|D|} \times \log_2 \frac{|D_j|}{|D|}$$

$$= \text{Split info}_{(age)} = -\sum \frac{|D_j|}{|D|}$$

$$= -\frac{5}{14}\log_2 \frac{5}{14} - \frac{4}{14}\log_2\left(\frac{4}{14}\right) - \frac{5}{14}\log_2\left(\frac{5}{14}\right)$$

$$= -\frac{5}{14}(-1.4854) - \frac{4}{14}(-1.8074) - \frac{5}{14}(-1.4854)$$

$$= 1.58$$

$$G_R = \frac{\text{Gain}}{\text{Split info}} = \frac{0.246}{1.58} = 0.156$$

$$\Rightarrow \text{Split info (income)} = -\sum \frac{|D_j|}{|D|}$$

$$= -\frac{4}{14}\log_2 \frac{4}{14} + \frac{6}{14}\log_2 \frac{6}{14} + \frac{4}{14}\log_2 \frac{4}{14}$$

$$= -\frac{4}{14}(-1.8074) - \frac{6}{14}(-1.2224) - \frac{4}{14}(-1.8074)$$

$$= 1.557$$

$$G_R = \frac{\text{Gain}}{\text{Split info}} = \frac{0.029}{1.557} = 0.019$$

$= 0.443$

$\Rightarrow$ D_1 {medium, High} D_2 {low}

$= 10/6/4$ $\frac{4}{3/1}$

gini (income) (m/H) $= \frac{10}{14}\left[1-\left[\frac{6}{10}\right]^2+\left[\frac{4}{10}\right]^2\right]+\frac{4}{14}\left[1-\left(\frac{3}{4}\right)^2+\left(\frac{1}{4}\right)^2\right]$

$= \frac{10}{14}\left[1-[0.36+0.16]\right]+\frac{4}{14}\left[1-0.563+0.063\right]$

$= \frac{10}{14}[0.48]+\frac{4}{14}[0.37]$

$= \frac{4.8}{14}+\frac{1.48}{14}$

$= \frac{6.28}{14}$

$= 0.449$

D_1 {Low, High} D_2 {medium}

$8 \rightarrow 5, 3$ $6 \rightarrow 4, 2$

Gini = 0.458

Gini index

D1	D2	value
LM	H	0.443
LH	M	0.458
MH	L	0.450

11/8/24